1 Abstract

Survival prognosis is still an integral component of personalized oncology, but because cancer is highly complex, there is a need to move from single-modality analysis to comprehensive multi-omics integration. This study presents an innovative deep learning framework aimed at improving pan-cancer survival prediction through the integration of diverse biological data, including genomic, transcriptomic, and DNA methylation data.

Our approach employs a convolution-based autoencoder architecture combined with an attention mechanism to extract and weight significant biological signals from different modalities. Unlike traditional statistical models, which often struggle with high dimensionality and non-linear relationships in biological systems, the proposed model effectively captures complex patterns in the data. Furthermore, by adopting a multimodal fusion strategy, the model captures cross-omic interactions that are frequently overlooked in late-stage integration approaches.

Preliminary evaluations demonstrate improved predictive performance, as indicated by the concordance index (C-index), while also identifying survival-associated subtypes across fifteen distinct cancer types. These findings highlight the potential of deep representation learning in integrating complex multi-omics data and providing meaningful clinical decision support, ultimately enabling more precise therapeutic planning.

2 Introduction

The rising incidence of cancer worldwide necessitates more reliable computational approaches for predicting patient outcomes and guiding personalized treatment strategies. Traditional prediction models have primarily relied on clinical variables such as age, tumor grade, and stage, or single-modality molecular data such as gene expression. However, cancer is a highly complex disease driven by interactions across multiple biological layers, including DNA, RNA, and proteins.

As a result, pan-cancer analysis, which involves the simultaneous study of multiple cancer types to identify shared molecular patterns, has emerged as an important research direction for discovering universal survival drivers and therapeutic targets.

Despite the availability of large-scale multi-omics datasets such as those provided by The Cancer Genome Atlas (TCGA), several significant challenges remain. First, integrating heterogeneous datasets often introduces noise and redundancy, which can degrade model performance without effective feature selection. Second, many deep learning models function as black boxes, limiting interpretability and reducing clinical trust. Third, high computational cost and the risk of overfitting on relatively small sample sizes hinder the practical application of these models in real-world clinical settings.

To address these challenges, this study proposes a deep learning-based multimodal integration framework. The model utilizes dedicated feature learning modules for each omics type and incorporates an attention-based fusion mechanism to identify the most informative biological signals. This approach not only improves survival prediction accuracy but also enhances the understanding of molecular heterogeneity across different cancer subtypes.

The ultimate goal of this research is to develop a scalable and efficient system that supports clinicians in early risk identification and enables personalized treatment planning.